

HyperGranular-RAG: Benefits and Boundaries of Structured Evidence Completion in Multi-Hop Question Answering

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Abstract

Evidence selection for multi-hop retrieval-augmented generation is a joint coverage problem constrained by context capacity. A relevant sentence can still be insufficient without a complementary bridge, yet introducing that bridge may displace useful context. This article examines structured evidence completion through HyperGranular-RAG: a geometry-based organization of candidate sentences, a seed-relative facet-hyperedge selector, and a bounded insertion policy that protects the beginning of a dense ranking. We consolidate the historical compact, cross-space sidecar, and BGE-native implementations while keeping their data boundaries and scoring definitions separate. With Qwen2.5-1.5B-Instruct, compact completion improves answer F1 over MiniLM Dense by 0.01478, 0.01140, and 0.01357 in three closed-candidate HotpotQA and MuSiQue evaluations; the recorded 95% paired bootstrap intervals exclude zero. The specified compact facet selector also exceeds a centroid-only alternative. In contrast, the original compact system falls below BGE, and neither tested BGE extension establishes an increment over that baseline. Generator transfer remains unresolved. We relate these outcomes to candidate placement, supporting-evidence turnover, and the distinction between group-level facets and prompt-visible coverage. Analytical results show why increased granular-ball compactness alone is not evidence of semantic improvement and why heuristic expansion cannot inherit an answer-quality guarantee from a coverage surrogate. The resulting account identifies a useful compact-backbone intervention and the empirical and theoretical limits of that intervention, without treating later synthetic prototypes as evaluated methods.

Keywords: retrieval-augmented generation; multi-hop question answering; evidence selection; granular representations; uncertainty.

1 Introduction

Retrieval-augmented question answering depends on two distinct capabilities: finding relevant information and supplying a generator with a usable combination of that information. A strong relevance ranking addresses the first capability, but does not necessarily satisfy the second. For a question requiring an entity bridge, several highly relevant sentences about the first entity may be less useful than one additional sentence connecting it to the second. Dense retrieval (Karpukhin et al., 2020) provides a natural backbone for this process, while multi-hop benchmarks expose the need to assemble related facts (Yang et al., 2018; Trivedi et al., 2022).

A finite context budget makes completion a replacement decision. If a system retains at most twenty evidence units, moving an additional candidate into the context can remove another unit. The new item may increase annotated support coverage without improving the answer; it may also duplicate retained evidence or move information to a position the generator uses poorly. Position-sensitive context use has been documented in language models (Liu et al., 2024). Consequently, the number of retrieved supporting facts, their placement, and the answer produced from them are related but non-interchangeable outcomes.

HyperGranular-RAG studies a bounded intervention on this ranking. Candidate sentence embeddings are organized into local granular balls. Query terms define facets over these balls, and seed-relative facet

hyperedges identify candidate regions that differ from the initial seeds. A placement policy retains the first ten dense entries and supplies at most four candidates before filling the remaining positions from the dense order. The method is an evidence-completion layer: it preserves a retrieval backbone and does not replace corpus search with a learned graph model.

This article asks four connected questions. First, can structured completion improve a compact dense retriever on end-to-end answers? Second, which of the available component comparisons support that improvement? Third, does the increment persist when the primary ranking is supplied by BGE? Fourth, what do evidence-turnover and geometric analyses explain about the observed boundary? These questions organize the presentation, rather than the chronology of implementation and evaluation.

The empirical answer is conditional. Across three Qwen evaluations, the historical compact implementation improves over MiniLM Dense. In the joint component study, the facet-conditioned selector exceeds the specified centroid-only control. Nevertheless, the original compact system is worse than BGE. Adding compact candidates to BGE and rebuilding the structure in BGE space both produce intervals crossing zero. The positive compact effects therefore establish a capability within that system, while the BGE results limit claims about general retrieval improvement.

The interpretation requires precise versioning. The historical compact selector computes scores once relative to the seeds, rather than recomputing true coverage marginals after every selection. Its recorded radius threshold does not independently trigger a split under the frozen size constraints. BGE-native reconstruction also changes geometry and scoring, so it is not an encoder-only substitution. Later engineering repairs and synthetic dynamic prototypes are not sources of the historical answer-quality results.

Our contribution is a complete, bounded empirical account of structured completion, accompanied by definitions that make its mechanisms testable. We provide the historical algorithms and their distinct comparisons; consolidate positive, negative, and unresolved answer-quality effects; connect candidate addition to evidence displacement; and derive the geometric and objective-function limits of possible explanations. The extended analysis uses existing results and mathematical reasoning, not new answer-quality experiments. It does not establish that granular balls are indispensable or that facet hyperedges outperform all simpler diversity and coverage rules.

2 Positioning within Retrieval and Evidence Selection

2.1 From Relevance Ranking to Connected Retrieval

Dense passage retrieval learns query and passage encoders for relevance matching (Karpukhin et al., 2020). Multi-hop retrieval introduces dependencies between retrieval steps. MDR conditions a subsequent retrieval on preceding evidence (Xiong et al., 2021), whereas IRCOT alternates retrieval with generated reasoning (Trivedi et al., 2023). Those methods change how a system reaches evidence. Our closed-candidate intervention instead changes which already available units enter a bounded context.

This distinction determines the scope of evaluation. A method can improve selection from a supplied candidate set without improving first-stage corpus recall. Conversely, a corpus retriever can reach the correct document without selecting the sentence the generator needs. The present study addresses the second type of problem conditional on candidate availability. It therefore does not claim to solve open-domain reachability.

2.2 Structure at Different Representational Levels

RAPTOR constructs a hierarchy through recursive abstraction (Sarathi et al., 2024). HippoRAG uses knowledge-graph structure and propagation (Gutiérrez et al., 2024). HyperGraphRAG encodes n-ary knowledge using hypergraph-structured representation (Luo et al., 2025). These methods differ in the

object being organized, how relations are produced, and how the structure is traversed. Their results cannot be imported as measurements under our generator and evidence budget.

In HyperGranular-RAG, a granular ball is a local group of sentence embeddings with a center and radius, inspired by granular representations (Xia et al., 2019). A facet hyperedge relates seed balls to an expansion ball through query-term coverage. This relation is not an asserted real-world fact and does not result from an entity-relation extraction model. The structure is query-local, computed from the available candidate set, and used to rank a small expansion. The historical study contains no budget-aligned direct comparison with RAPTOR, HippoRAG, or HyperGraphRAG.

2.3 Coverage and Diversity as Alternative Explanations

A structured selector may improve a ranking simply by avoiding redundant evidence. MMR is a classical relevance–redundancy rule (Carbonell and Goldstein, 1998). Explicit coverage and diversity objectives can offer another route to evidence selection (Lin and Bilmes, 2011). These alternatives matter because a useful grouping heuristic does not by itself establish that high-order relations caused the benefit.

We use the term *facet-conditioned* to describe the implemented rule, not to settle that causal question. Its compact NoFacet comparison replaces the selector with centroid-only ball expansion. It does not hold every possible coverage or diversity principle constant, and it does not compare the method to a sentence-level coverage optimizer. The distinction between a pipeline comparison and a pure component intervention is maintained throughout the analysis.

2.4 Context Use and Empirical Uncertainty

Long-context evaluations show that evidence location can affect model performance (Liu et al., 2024). This motivates protecting an initial retrieval prefix, but does not prove that ten entries is an optimal protection budget. Likewise, statistical testing guidance emphasizes the relationship between a statistic, a null hypothesis, and the dependence structure of the sample (Dror et al., 2018). We report historical estimates with their recorded intervals and describe the evaluator and inferential caveats explicitly, rather than strengthening a result through terminology.

3 Problem Definition and Historical Algorithms

3.1 Candidate, Ranking, and Prompt Spaces

Let q denote a question and U_q the finite set of sentence units supplied by its benchmark context. A unit includes a document title, sentence text, and a stable identity. The encoder receives the title and sentence, and produces a normalized embedding $\hat{e}_u$. With query embedding $\hat{e}_q$, the dense relevance score is

$$s(q, u) = \hat{e}_q^\top \hat{e}_u. \quad (1)$$

Units are ordered by decreasing score and increasing identity. The baseline list D_q contains the first $K_q = \min(20, |U_q|)$ unique units. The completed list R_q has the same length.

Three evidence objects must be distinguished. The *candidate insertion set* contains units accepted by an expansion policy. The *final ranking* is the ordered list after placement, deduplication, and truncation to K_q . The *prompt-visible evidence* is the text actually retained by the token-limited prompt. Equality at one level does not automatically imply equality at the next. In particular, a fixed number of sentence units is not a fixed number of tokens.

3.2 Compact Granular-Ball Organization

The compact method uses MiniLM embeddings for both relevance and grouping. It begins with the whole candidate set as one root. For a ball B ,

$$c_B = \text{norm} \left(\sum_{u \in B} \hat{e}_u \right), \quad d(u, B) = 1 - \hat{e}_u^\top c_B, \quad r_B = \max_{u \in B} d(u, B). \quad (2)$$

Compactness is $C(B) = 1 - |B|^{-1} \sum_{u \in B} d(u, B)$. The historical split eligibility rule is

$$\text{depth}(B) < 6 \quad \wedge \quad |B| \geq 4 \quad \wedge \quad (|B| > 3 \vee r_B > 0.78). \quad (3)$$

Because $|B| \geq 4$ implies $|B| > 3$, the radius disjunction is inactive. The first split seed is farthest from the center; the second has minimum cosine similarity to the first. Assignment uses the more similar seed and the frozen tie rule. A split is accepted only when each child has at least two members. Otherwise the unsplit group becomes a leaf, possibly with more than three members.

Thus the historical partition adapts to size and geometry, but is not independently radius-triggered. We retain its actual definition rather than replacing the predicate and attributing old results to a revised algorithm. The compactness analysis in Section 4 explains why geometrically more compact children are not, on their own, validated semantic evidence units.

3.3 Facet Hyperedges and Static Selection

The fixed tokenizer extracts ASCII-alphanumeric content terms and removes the recorded stop words. Let T_q denote query terms and $T_B = T_q \cap \bigcup_{u \in B} T(u)$ the facets of ball B , using its title and sentence text. The two balls with highest query-centroid similarity are the seeds. Their union of facets, T_{seed} , defines the reference for expansion.

For a non-seed ball, write $n_B = |T_B \setminus T_{\text{seed}}|$, $t_B = |T_B|$, $m = |T_q|$, $b_B = \hat{e}_q^\top c_B$, $a_B = \max_{S \in \text{seeds}} c_B^\top c_S$, and $\rho_B = |T_B \cap T_{\text{seed}}|/|T_B|$ when T_B is nonempty. The candidate score is

$$h_B = .45 \frac{n_B}{\max(m, 1)} + .20 \frac{t_B}{\max(m, 1)} + .20b_B + .15(1 - a_B) - .20\rho_B - .05|B|/8. \quad (4)$$

Eligibility requires $n_B \geq 1$, $|B| \leq 8$, $b_B \geq .12$ or $a_B \geq .05$, $|B|/n_B \leq 6$, $\rho_B \leq .90$, and $h_B \geq .10$. Eligible balls are ordered by decreasing h_B and stable edge identity; at most two distinct expansion balls are selected. Their units retain dense-score and identity order. The historical unit-level facet bonus is zero.

A facet hyperedge is the relation between the seed balls and an eligible expansion ball contributing seed-relative new facets. It identifies a region of candidates; it does not assert that every unit in that region supplies the missing fact. All eligibility and score calculations occur once. Updating an internal covered-term set after accepting a ball does not recompute the remaining scores. This static rule is different from true marginal-coverage greedy selection.

3.4 Protected Bounded Insertion

Let P_q be the first $\min(10, K_q)$ entries of D_q . The compact method traverses the expansion order, skips duplicates in P_q , and accepts at most four units with dense score at least 0.1957079917192459. The floor is a historical development-calibrated 25th percentile, transferred unchanged. It is not a percentile recomputed on each evaluation set. The policy concatenates the protected prefix, accepted units, and remaining dense entries, applies stable deduplication, and truncates to K_q .

An accepted unit can already be in dense positions 11–20. Such an event is a promotion within the baseline set; it differs from introducing a new unit outside that set. Both consume placement capacity. This distinction matters when interpreting insertion counts as evidence expansion. Candidate counts cannot be substituted for counts of genuinely new supporting facts.

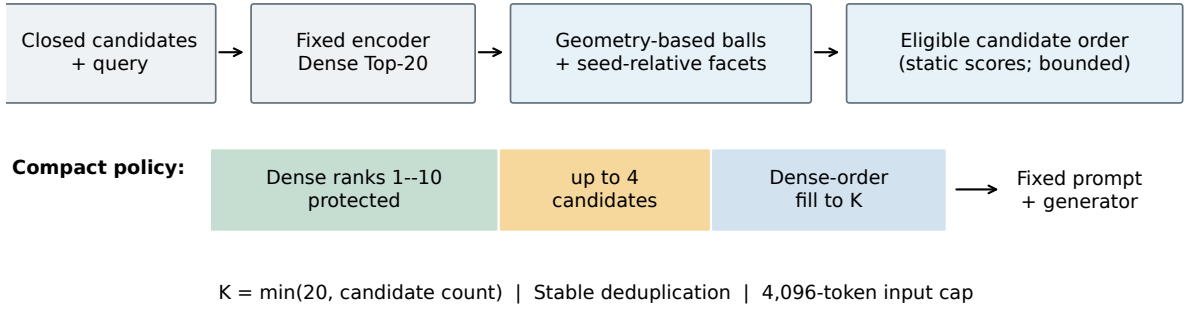


Figure 1: Historical compact completion. The root includes the full candidate pool. Static facet selection precedes placement; final list length and the generator’s token cap are separate constraints. This diagram depicts the historical implementation, not the unexecuted dynamic prototype.

3.5 Cross-Space Sidecar

The sidecar holds BGE’s dense ranking as the reference while the frozen MiniLM structure proposes eligible units outside BGE Top-20. It introduces no learned score fusion. Protected placement inserts its accepted candidates after the BGE prefix; unprotected placement puts the same candidates first. Both then fill from BGE and enforce the same effective list length.

The sidecar asks whether a compact structure contains useful complementary evidence that BGE does not already expose. It is more targeted than comparing the original MiniLM pipeline against BGE, because the primary ranking now comes from BGE in both arms. It remains a cross-space construction: the candidate source and its score geometry are not native to BGE.

3.6 BGE-Native Reconstruction

The native implementation changes that geometry. A separate development family crosses leaf target $\{4, 6\}$, insertion budget $\{2, 4\}$, unit-score percentile floor $\{.50, .75\}$, and broad/strict facet gates, giving 16 configurations. The selected C10 uses leaf target six, budget four, floor .50, and broad gates. The confirmation comparison fixes that choice.

Native splits use depth limit eight, minimum child fraction .20, and a weighted-child compactness improvement greater than 10^{-10} . Seed selection uses deterministic center-relative two-seed geometry and identity-based ties. Within a query, unit relevance, ball relevance, radius, compactness, and density are converted to empirical percentiles, where the percentile of a value is the fraction of values less than or equal to it. Density is group size divided by radius with the recorded small denominator floor.

Two seed balls are selected by relevance percentile, density percentile, and identity. Broad gates require ball-relevance percentile at least .35 and density percentile at least .25; strict gates use .50 for both. Eligible expansion balls must have a seed-relative new facet and are ranked by

$$h_B^N = .55p_B + .25p_{\rho,B} + .15p_{C,B} + .05 \min(n_B, 3)/3. \quad (5)$$

At most four balls supply units. A candidate must be outside BGE Top-20 and meet the configured unit percentile floor. Its score is

$$h_u^N = .45p_u + .25p_B + .15p_{\rho,B} + .10p_{C,B} + .05 \min(n_B, 3)/3. \quad (6)$$

Native NoFacet removes the new-facet gate and uses $.50p_u + .30p_B + .10p_{\rho,B} + .10p_{C,B}$ for units. It therefore changes several coefficients as well as facet eligibility. Its result is a pipeline contrast, not a coefficient-preserving pure ablation. The native study cannot be read as changing only the encoder in the compact algorithm.

4 Analytical Properties and Their Limits

4.1 Compactness Is a Resultant-Length Statistic

The following properties concern the mathematical representation. They are not claims about new empirical performance.

Proposition 1. *For nonempty B with unit embeddings x_i , and nonzero $s_B = \sum_{i \in B} x_i$, the historical mean-cosine compactness satisfies*

$$C(B) = \frac{1}{|B|} \sum_{i \in B} x_i^\top \frac{s_B}{\|s_B\|} = \frac{\|s_B\|}{|B|} \in [0, 1]. \quad (7)$$

The scalar has the continuous extension $C(B) = 0$ at $s_B = 0$.

Proof. Linearity of the inner product gives $\sum_i x_i^\top s_B / \|s_B\| = s_B^\top s_B / \|s_B\| = \|s_B\|$. The upper bound follows from $\|\sum_i x_i\| \leq \sum_i \|x_i\| = |B|$. Continuity follows from continuity of the norm. $\square$

The scalar extension does not determine a normalized center at zero. The compact implementation’s zero-vector convention and the native implementation’s fallback direction can consequently have different assignment behavior even though the exact zero-sum scalar is the same.

Proposition 2. *For every partition $B = \bigsqcup_{j=1}^m B_j$ into nonempty groups,*

$$\sum_j \frac{|B_j|}{|B|} C(B_j) \geq C(B). \quad (8)$$

Proof. Writing $s_j = \sum_{i \in B_j} x_i$, the left side is $|B|^{-1} \sum_j \|s_j\|$, while the right side is $|B|^{-1} \|\sum_j s_j\|$. The triangle inequality gives the claim. $\square$

Equality holds when the nonzero group sums share a direction. Otherwise a strict increase can arise without any semantic alignment. For four orthogonal unit vectors, parent compactness is $1/2$; every partition into two pairs has weighted compactness $1/\sqrt{2}$. A positive compactness change is therefore not evidence that a split recovered a reasoning relation. The native balance and depth rules constrain a procedure, but do not change this logical limitation.

4.2 Scalar Stability Does Not Guarantee Partition Stability

For corresponding perturbed vectors x'_i , the reverse triangle inequality yields

$$|C'(B) - C(B)| \leq \frac{\|\sum_i (x'_i - x_i)\|}{|B|} \leq \frac{1}{|B|} \sum_i \|x'_i - x_i\|. \quad (9)$$

This is a stability bound for the scalar with membership fixed. It does not bound changes in the normalized center near a zero sum, nor does it keep discrete assignments fixed. For example, $x_1 = (1, 0)$ and $x_2 = (-1, \epsilon)/\sqrt{1 + \epsilon^2}$ have a sum whose direction approaches the positive vertical axis as $\epsilon \downarrow 0$. Replacing ϵ with $-\epsilon$ makes the input perturbation arbitrarily small while reversing that limiting direction. A stability claim for a full partition would need additional separation assumptions.

4.3 Seed-Relative Novelty Is Not a Dynamic Marginal

Suppose seeds cover $\{\alpha\}$ and three candidate groups cover $\{\beta\}$, $\{\beta\}$, and $\{\gamma\}$. A static score order can select the first two. Both were new relative to the seeds, but the second adds no facet after the first is selected. A dynamic coverage rule would reconsider the remaining candidates at that point. This is a structural difference between algorithms, not an estimate of how often the event occurs in the historical corpus.

Likewise, let a selected group contain two sentences with facets $\{\alpha\}$ and $\{\beta\}$, while the placement rule emits only the first. The group union is $\{\alpha, \beta\}$ but selected-unit coverage is $\{\alpha\}$. We call their difference *phantom coverage*: a facet credited to the group but absent from the selected units. Prompt truncation can further reduce visible content. None of the historical aggregate results measures the prevalence or answer effect of this particular discrepancy.

4.4 A Coverage Guarantee Applies Only to Its Own Objective

To make the distinction concrete, consider a fixed set of units, fixed observable facets $T(u)$, and nonnegative weights w_t . Define

$$F(S) = \sum_t w_t \mathbf{1} \left[t \in \bigcup_{u \in S} T(u) \right]. \quad (10)$$

For a fixed protected set P , let $f_P(A) = F(P \cup A) - F(P)$. The residual objective is normalized, monotone, and submodular: as A grows, the set of new facets available from any additional unit can only shrink. This kind of explicitly specified coverage objective is consistent with the objective-based approach discussed by [Lin and Bilmes \(2011\)](#).

If at most b additional units are allowed and greedy selection recomputes the true marginal each step, then for an optimal additional set O ,

$$f_P(O) - f_P(A_i) \leq \sum_{u \in O \setminus A_i} [f_P(A_i \cup \{u\}) - f_P(A_i)] \leq b[f_P(A_{i+1}) - f_P(A_i)]. \quad (11)$$

Iterating gives $f_P(A_b) \geq [1 - (1 - 1/b)^b]f_P(O) \geq (1 - 1/e)f_P(O)$ for $b \geq 1$. The $b = 0$ case is trivial, and selecting all remaining units is optimal if fewer than b are available.

This is the standard greedy coverage bound, specialized to an explicitly defined surrogate under a cardinality budget and a fixed prefix. The historical static score in Equation 4 does not implement that algorithm. Variable token costs, negative penalties, changing facets, and generated answer F1 are outside the statement. In particular, a coverage guarantee cannot be used as an answer-quality guarantee.

4.5 Complementarity Can Defeat a Greedy Interpretation

A two-group activation objective already illustrates the difficulty. For $e = \{a, b\}$, let $H(S) = \mathbf{1}[|S \cap e| \geq 2]$. Adding b to the empty set has gain zero, while adding it to $\{a\}$ has gain one, violating diminishing returns. Complementarity can be desirable without being submodular.

Degree normalization presents a separate issue. Take hyperedges $\{a, b\}$ and $\{b, c\}$, and divide newly activated edges by the incoming group's degree. The path $a \rightarrow b$ accumulates $1/2$, whereas $b \rightarrow a$ accumulates 1 for the same final set $\{a, b\}$. Such scores cannot be exact marginals of an order-independent set potential. These examples delimit potential extensions; they are not the scoring rule of historical compact q25 and supply no new performance result. A bounded lookahead alternative would require its own objective and cost evaluation.

Evaluation	HotpotQA	MuSiQue	Scientific role
Initial compact	1,000	3,000	Two dataset-specific evaluations of frozen compact completion
Joint component	1,000	1,500	Seven-arm compact, lexical, hybrid, strong-dense, and component comparison
Cross-space sidecar	1,000	1,500	Four-arm comparison around BGE using compact candidates
Native development	500	750	Selection from a fixed 16-configuration family
Native confirmation	1,000	1,500	Evaluation of the selected configuration and fixed comparators
Generator transfer	1,000	3,000	Initial rankings reused with Gemma mobile-QAT

Table 1: Data boundaries. Counts are questions, not retrieval units. All come from training releases held out within the project, rather than official hidden benchmark tests. Development and confirmation are never pooled.

5 Empirical Design

5.1 Data Boundaries and Research Questions

HotpotQA provides multi-hop questions with sentence-level supporting facts (Yang et al., 2018). MuSiQue constructs connected questions from single-hop components, with an answerable subset and a separate setting including unanswerable instances (Trivedi et al., 2022). We use the answerable training release, not the unanswerable extension. Both benchmarks provide the closed candidate contexts used in this study.

Table 1 separates the evaluations. The initial compact studies test one frozen policy on two datasets. The joint component set tests the same compact pipeline against additional baselines and component alternatives. The sidecar and native confirmation sets test different interventions around BGE. Their manifests record query-identity separation from earlier evaluation sets; the native confirmation also excludes its development sample. Generator transfer deliberately reuses the initial rankings and questions. Thus it is not an independent replication of candidate selection.

The experimental questions are whether compact completion improves answers, which specified component comparisons account for its behavior, whether the benefit extends to BGE, and whether it persists for an additional generator. Post-decision evidence-turnover analyses address interpretation, not additional confirmation hypotheses.

5.2 Baselines and Component Comparisons

The compact baseline uses `sentence-transformers/all-MiniLM-L6-v2`. The strong baseline is `BAAI/bge-large-en-v1.5`, with a model revision recorded in the implementation manifest. The component evaluation also contains BM25 and a fixed Dense-BM25 hybrid. These comparisons hold the candidate boundary and generator fixed within an evaluation.

Compact NoFacet preserves granular-ball geometry but selects up to two eligible expansion balls by centroid relevance. NoProtection alters placement. The sidecar and native studies each contain BGE, Protected, NoFacet, and Unprotected arms. Native NoFacet changes both facet eligibility and scoring coefficients, as defined above. These are the implemented controls, not interchangeable labels for a universal facet ablation.

A flat-unit granular-ball control was not completed because a unique matched mapping of ball seeds, split rules, and group budgets was not specified. That omission cannot establish either the necessity or the uselessness of granular balls. Random grouping, matched-count ordinary clustering, MMR, and sentence-level coverage were also not formal arms in these historical answer-quality evaluations.

5.3 Generation and Development Choices

Qwen2.5-1.5B-Instruct FP16 is the main generator. All main comparisons use a fixed short-answer evidence prompt, a 4,096-token input cap, greedy decoding, batch size one, and a maximum of 32 newly generated tokens. The instruction asks for the shortest answer using only the supplied evidence and UNKNOWN if it is insufficient. Outputs are stripped of leading and trailing whitespace. The recorded hardware is an RTX 4060 Laptop with 8GB VRAM.

An earlier 200-question generator-selection development study selected this Qwen deployment. The additional Gemma 4 E2B configuration runs in the official mobile-QAT format. The comparison is conditioned on hardware, prompt, cap, and deployable numerical format. It is not evidence that Qwen’s architecture is universally stronger or more efficient.

Native C10 was selected using a separate development evaluation, whose equal-weight answer-F1 difference was $+0.003143$. Its configuration was then fixed for confirmation. The development value is reported as selection context only; it is not averaged with the confirmation effect and does not override it. The pre-specified native search ended after confirmation.

5.4 Answer Scoring and Support Granularity

For each question, normalized answer exact match and token-overlap F1 measure the generated answer. Multiple reference aliases are maximized where the historical dataset scorer specifies them. Mean answer F1 is the primary endpoint; EM provides a stricter exact-answer view.

The scoring lineage is material. The first HotpotQA implementation includes the official special-answer exception: an unequal prediction involving yes/no/noanswer receives zero F1. The initial MuSiQue implementation uses its frozen answer normalization and aliases. The generator-transfer scorer keeps the HotpotQA-specific exception. In contrast, the component, sidecar, and native implementations use alias-max normalized token overlap without that exception. Token overlap can give nonzero credit to a longer phrase containing *yes* against the reference *yes*, whereas the special rule assigns zero if their normalized answers differ.

We preserve each historical result with its actual scoring implementation. The later rows are marked *legacy scoring*, not represented as a newly verified official HotpotQA metric. The prevalence and aggregate impact of special-answer mismatches in those predictions remain unquantified. No rescoring has been carried out for this article.

For supporting evidence, let G_q be the dataset’s annotated support identities and $V_q(R)$ those covered by ranking R . Then

$$\text{CR@20}(q) = \mathbf{1}[G_q \subseteq V_q(R_q)], \quad \text{ER@20}(q) = \frac{|G_q \cap V_q(R_q)|}{|G_q|}. \quad (12)$$

HotpotQA support identities are sentence-level. MuSiQue supplies supporting paragraphs, and coverage follows that annotation granularity. It is not MuSiQue sentence-level or complete-chain verification. These retrieval metrics concern ranked evidence before the generator’s use of the context.

5.5 Paired Estimation and Historical Inference

For methods a and b , question i , and dataset d , write $\delta_{di} = F1(a, di) - F1(b, di)$. A dataset-specific effect is $\hat{\Delta}_d = n_d^{-1} \sum_i \delta_{di}$. Joint studies target

$$\hat{\Delta}_{\text{eq}} = \frac{1}{2} \hat{\Delta}_{\text{HotpotQA}} + \frac{1}{2} \hat{\Delta}_{\text{MuSiQue}}. \quad (13)$$

This weights datasets equally rather than weighting all questions equally. The recorded procedure uses 10,000 paired query bootstrap samples; joint comparisons resample within each dataset and combine

Setting and method	F1	EM	CR@20	ER@20
HotpotQA / Dense	.42150	.35600	.73700	.87860
HotpotQA / Full	.43628	.36600	.79800	.90818
MuSiQue / Dense	.13595	.10033	.58900	.80553
MuSiQue / Full	.14735	.11033	.65000	.84128
Joint / Dense [†]	.30446	.25467	.66617	.84529
Joint / Full [†]	.31803	.26400	.72217	.87657

Table 2: Compact MiniLM with Qwen. Joint metrics are dataset-equal-weighted. All values are on the 0–1 scale. HotpotQA and MuSiQue support coverage use different annotation granularities. [†]Legacy scoring; scores are not silently harmonized across studies.

the two means. We display the original percentile intervals, with no pooling across different evaluation boundaries and no replacement of the original bootstrap.

These intervals describe a query-resampling model conditional on the selected pipeline. They do not integrate uncertainty over configuration selection, pretraining, or alternate model initialization. Query-disjoint sets can still share source documents, and document-level dependence is not captured by the query bootstrap. Sample sizes reflect the resources allocated to the study, not a formal prospective power guarantee.

Some historical decision files label the nonpositive fraction of uncentered bootstrap differences as a one-sided p-value, followed by Holm adjustment for a specified family. A tail fraction computed around the observed empirical distribution does not automatically supply a null-calibrated p-value. Following the distinction between effect estimation and hypothesis testing (Dror et al., 2018), we retain those files as historical decisions but do not present their adjusted values as a newly validated familywise test. The manuscript reports effect estimates and recorded intervals, and describes whether those intervals exclude or cross zero. No equivalence margin was pre-specified, so an unresolved comparison is not an equivalence result.

5.6 Evidence Separation and Verification Scope

Confirmation answers and supporting-fact labels were separated from retrieval, prompt construction, and generation until their outputs were fixed. Designated development labels could select configurations; therefore, label-free retrieval does not mean that no development labels were used anywhere in the research process. Post-decision support audits were not used to retune the corresponding completed evaluation.

The repository preserves ranking and prediction records, configuration manifests, determinism checks, and historical verification outputs. Verification coverage must be interpreted per computation. Artifact identity checks establish file identity; they do not establish correct scientific definitions. A separately named verifier may share semantic dependencies. Later repairs to independent grouping or final metric reconstruction are not retroactive validation of older numerical outputs. This article performs aggregate evidence reconciliation for writing, rather than rerunning the historical experiments.

6 Answer-Quality Results

6.1 Repeated Compact-Backbone Improvements

Table 2 reports the compact results on their original scale. The initial HotpotQA effect is +0.01478 with interval [0.00020, 0.02988], and the MuSiQue effect is +0.01140 [0.00450, 0.01835]. The separate joint component study gives +0.01357 [0.00491, 0.02233]. Every recorded interval lies above zero, although the initial HotpotQA lower bound is close to it. Together, the three comparisons support a repeated benefit for this compact pipeline under their recorded metrics.

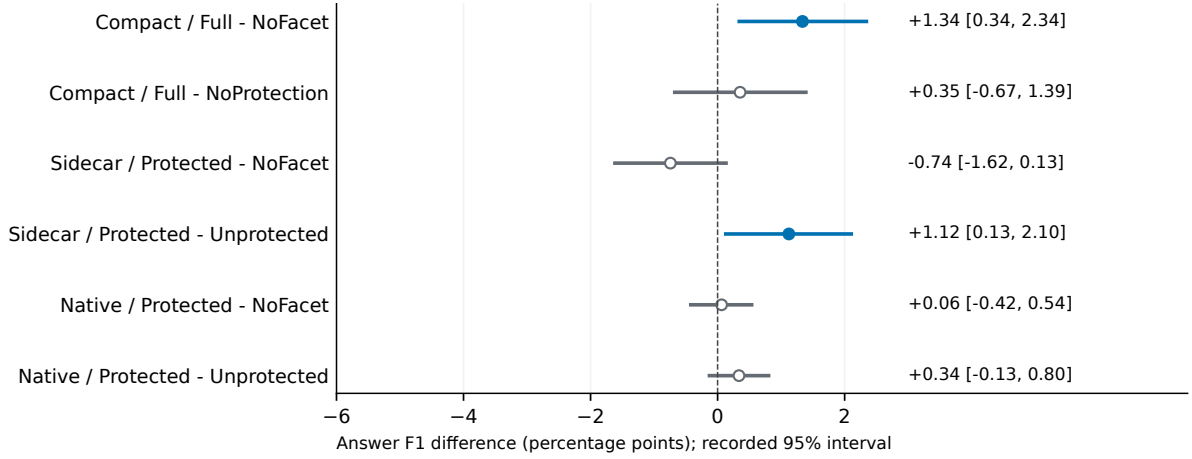


Figure 2: Component and placement effects in percentage points, with recorded 95% intervals. These are legacy-scoring results. Open markers cross zero. Native NoFacet changes scoring coefficients in addition to facet eligibility, so it is a pipeline comparison. The fixed candidate insertion sets in placement comparisons do not establish identical prompt-visible evidence.

The gain is not uniform over questions. The existing descriptive reconstruction counts 47 positive and 34 negative answer-F1 changes in the initial HotpotQA set, 125/77 in MuSiQue, and 102/73 in the joint set. These counts explain why a positive average is compatible with individual harms. They are answer-change counts, not counts of additional supporting evidence; the latter use a different event definition.

The mean placed-candidate counts are 1.9300, 2.4110, and 2.1628 for these three settings. The joint insertion mean and event counts are query-weighted over 2,500 questions, whereas the joint answer metrics in Table 2 are dataset-equal-weighted. This distinction prevents a larger dataset from implicitly changing the primary effect’s meaning.

6.2 Facet and Protection Comparisons

Compact FULL minus NoFacet is $+0.01336$ $[0.00341, 0.02343]$. The interval lies above zero for the specific centroid-only comparator. It supports an incremental role for facet-conditioned selection in that implementation. It does not isolate the intrinsic value of a hypergraph representation from all coverage or diversity alternatives. The compact NoProtection contrast is $+0.00354$ $[-0.00675, 0.01389]$, which remains unresolved.

The cross-space facet contrast is -0.00743 $[-0.01616, 0.00132]$, and its placement contrast is $+0.01122$ $[0.00129, 0.02104]$. Only the latter interval excludes zero. In native space, the facet pipeline contrast is $+0.00063$ $[-0.00422, 0.00536]$, while the placement contrast is $+0.00337$ $[-0.00129, 0.00803]$; both cross zero. Figure 2 displays all six contrasts to avoid transporting the compact facet result into an unsupported BGE claim.

6.3 Original Completion against a Stronger Baseline

In the joint component set, BGE has answer F1 .35801 compared with .31803 for original FULL. The paired FULL-minus-BGE difference is -0.03998 $[-0.05393, -0.02621]$, with its entire recorded interval below zero. The original MiniLM-based system is therefore worse than BGE under that evaluation. The gain over MiniLM cannot be substituted for an improvement over a stronger retrieval backbone.

The same set gives FULL-minus-BM25 -0.00755 $[-0.02259, 0.00773]$ and FULL-minus-hybrid -0.00320 $[-0.01591, 0.00942]$. Both are unresolved. Including these reference points shows that the compact gain is not a general win over all reasonable baselines.

Evaluation	Method	F1	EM	CR@20	ER@20
Joint component	BGE	.35801	.30467	.85400	.93902
	Original Full	.31803	.26400	.72217	.87657
Cross-space	BGE	.34301	.28000	.84967	.93594
	Protected	.34045	.27767	.84350	.93459
Native confirmation	BGE	.34569	.28500	.86167	.94288
	Protected	.34264	.28133	.86233	.94288

Table 3: Absolute strong-backbone results under legacy scoring, averaged equally across datasets within each evaluation. Different evaluation sets are not pooled. The identical rounded native ER@20 values do not imply identity of evidence sets.

6.4 Cross-Space and Native BGE Extensions

The sidecar tests complementarity while preserving BGE as the primary ranker. Its Protected-minus-BGE effect is -0.00256 $[-0.00998, 0.00458]$. The native reconstruction yields -0.00305 $[-0.00688, 0.00063]$ on its own confirmation set. Neither establishes an increment. Their small negative point estimates and intervals crossing zero do not demonstrate equality or prove that every BGE-compatible structural method would fail.

The native dataset-specific F1 differences are as follows:

$$\begin{aligned} \text{HotpotQA:} & \quad - .002907 \quad [-.008682, .002652], \\ \text{MuSiQue:} & \quad - .003184 \quad [-.008486, .001863]. \end{aligned}$$

The equal-weight EM difference is -0.003667 , with an interval of $[-.007667, .000004]$. These values corroborate the direction of the reported point estimates but do not resolve either dataset-specific comparison. The positive development value used to choose C10 is not confirmation evidence.

The unprotected policies are informative additional references. Sidecar Unprotected minus BGE is -0.01379 $[-0.02440, -0.00339]$; native Unprotected minus BGE is -0.00641 $[-0.01171, -0.00118]$. Both intervals lie below zero. Protection can reduce harm from an intervention even when the protected intervention does not exceed the original BGE baseline.

6.5 Generator Transfer

The pre-specified Gemma mobile-QAT deployment gives an equal-weight FULL-minus-Dense F1 effect of $+0.00516$ $[-0.00262, 0.01295]$. This result is unresolved. It is one additional generator configuration, not evidence of universal transfer or an intrinsic architecture ranking. Its reuse of initial rankings preserves the retrieval comparison, but does not add new candidate-selection boundaries.

7 Evidence Turnover, Cases, and Cost

7.1 Candidate Placement Is Not Uniform across Systems

The existing Gold-free descriptive trace summaries show different intervention rates. Under dataset-equal weighting, zero candidates are placed for 30.67% of compact queries, 41.43% of sidecar queries, and 80.80% of native queries. Four candidates are placed for 33.95%, 17.80%, and 9.32%, respectively (Figure 4). These percentages describe distinct query sets and distinct candidate rules; they are not the results of a matched intervention-rate experiment.

The sparse native intervention gives context to its small aggregate effect. Many queries preserve the original BGE ranking, while the modified queries still need candidates that improve the answer. A lower intervention rate can reduce both gain and harm. The trace does not establish whether the native gates are

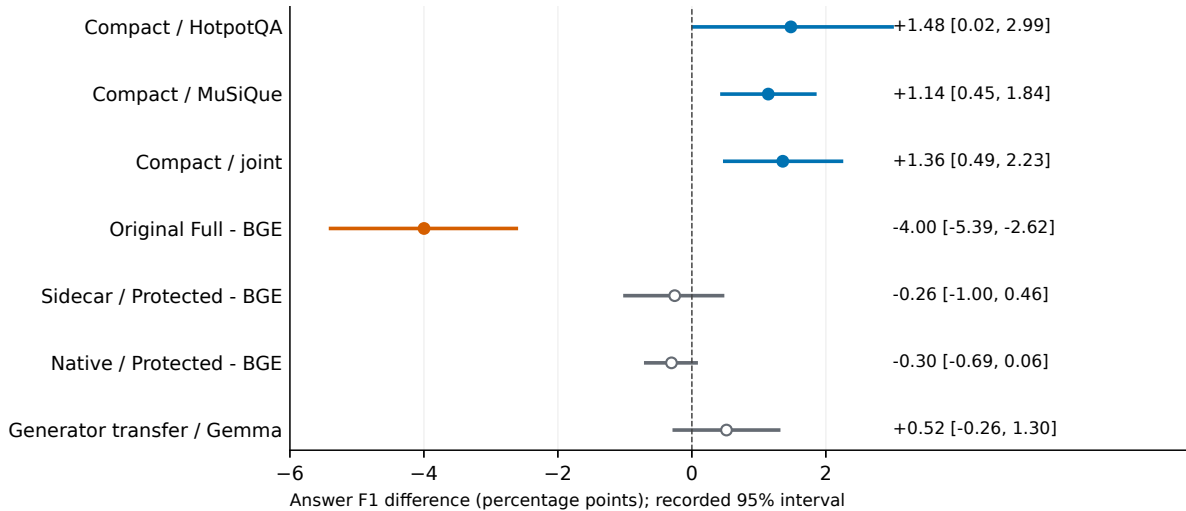


Figure 3: Main paired effects on a common percentage-point display scale. The three compact rows, original-versus-BGE, two BGE extensions, and generator transfer remain separate estimands. Lines are recorded 95% percentile intervals. This is not a fitted relationship between retriever strength and benefit.

Variant	Dataset	Questions	Added	Displaced	Net
Sidecar	HotpotQA	1,000	21	29	-8
Sidecar	MuSiQue	1,500	59	50	+9
Native	HotpotQA	1,000	3	2	+1
Native	MuSiQue	1,500	16	15	+1

Table 4: Existing post-decision supporting-evidence events for Protected relative to BGE. Counts use each dataset’s support identities, so a MuSiQue supporting paragraph is not a HotpotQA supporting sentence. These are descriptive totals, not causal effect estimates or new confirmation tests.

too strict, whether available candidates are unhelpful, or whether the generator fails to use them. Relaxing those gates after observing confirmation would be a new selection decision, not further analysis of the same frozen method.

7.2 Supporting-Evidence Addition and Displacement

The post-decision audits compare support identities entering and leaving the final ranking. Sidecar HotpotQA has 21 added and 29 displaced supporting units, for net -8 . Sidecar MuSiQue has 59 added and 50 displaced, for net $+9$. Native HotpotQA has 3 added and 2 displaced, while native MuSiQue has 16 added and 15 displaced, giving net $+1$ in each. Table 4 preserves these totals and their denominators.

These counts do not yield a monotone account of answer utility. The native system adds slightly more supporting evidence than it removes, yet both dataset-specific answer-F1 point estimates are negative. An added supporting unit may be redundant with retained text, and a displaced unit may have a different role from an added one. Token truncation and context order can also change what the generator effectively uses. The aggregate audits do not estimate the separate contributions of those mechanisms.

The practical implication is to report four objects separately: group-level facet coverage, selected-unit coverage, prompt-visible evidence, and answers. Only the last supplies the present primary endpoint. Treating an improvement in the first as a guarantee about the last would overlook both displacement and generation.

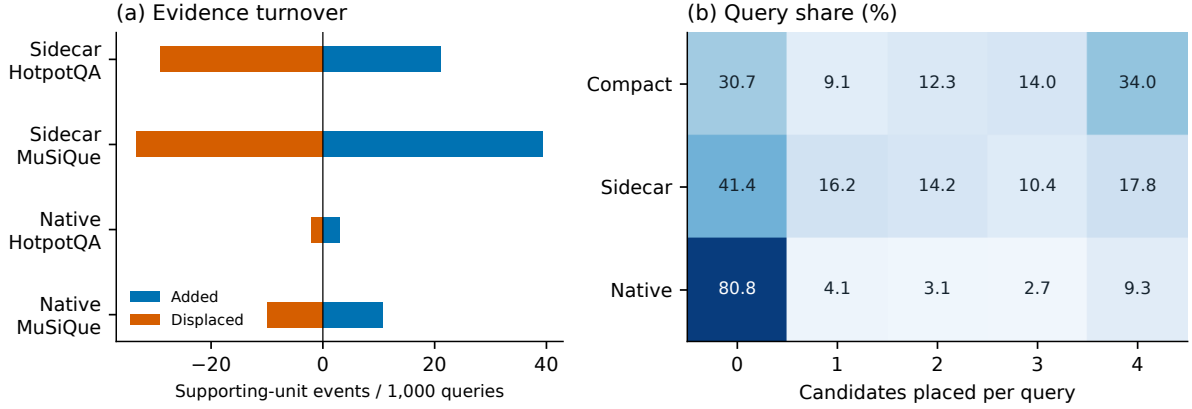


Figure 4: Existing descriptive mechanism summaries. Left: added and displaced annotated evidence per 1,000 questions, with direction represented by sign. Right: dataset-equal-weight query percentages by placed-candidate count. Candidate placement can include within-Top-20 promotion in the compact variant; it is not interchangeable with genuinely new evidence.

Case	Evidence relation	Answer transition
Compact success	Existing Khmer Rouge film description plus an added director–film bridge	Dense: UNKNOWN → Full: The Killing Fields; F1 0 → 1
Cross-backbone failure	BGE marital relation omitted by original compact Full; unrelated society sentence added	BGE: Sir Winston Churchill → original Full: UNKNOWN; F1 1 → 0

Table 5: Paraphrased illustrations from the existing representative-case records. The second case is not a protected-placement ablation. Original questions, evidence text, method identities, and selection rules remain in the shared source CSV.

7.3 Representative Success and Failure

The existing qualitative selection uses the example closest to the median answer-F1 change within a specified event family, with a stable identity tie-break. The success family requires a positive compact Full-minus-Dense answer change, improved complete recall, and an inserted supporting unit. The failure family requires a negative original-Full-minus-BGE answer change and a displaced supporting unit. This procedure selects illustrations after the outcome; it does not create an unbiased estimate of prevalence.

The success question asks which film by the director of *You and I* concerns the Khmer Rouge. Dense retrieves a description of *The Killing Fields* at rank nine but answers UNKNOWN. Completion adds, at rank twelve, a sentence about *You and I* that identifies Roland Joffé and connects him with *The Killing Fields*. The generated answer changes to the reference film, with F1 moving from zero to one. The added bridge makes the answer derivable from the displayed evidence, although one example cannot isolate a general causal mechanism.

The failure question asks whom the chairman of the Red Cross Aid to Russia Fund was married to. BGE retains a rank-two sentence identifying Clementine Churchill as Winston Churchill’s wife and answers correctly. The original compact Full ranking omits that supporting sentence and produces UNKNOWN; its added sentence about a different Red Cross society does not repair the link. This is a comparison of different retrieval pipelines, not an instance of the protected sidecar evicting BGE’s protected rank-two unit.

Recorded item	Value	Interpretation
Retrieval reconstruction	8.544 s	Cached native transaction
BGE generation	887.650 s	Per-method aggregate
Protected native generation	884.198 s	Per-method aggregate
Peak GPU memory	3.592 GiB	Recorded hardware
Embedding cache	670,420,228 bytes	Existing cache identity
Main transaction generation calls	10,000	Four arms $\times$ 2,500 queries
Generated in final recorded process	2,000	Resume-process counter
Recorded failed calls	0	Historical telemetry

Table 6: Native confirmation cost record. Transaction-wide and final-process counters are different fields; the table does not imply that 10,000 calls occurred in the final resumed process. No generation was run for manuscript preparation.

7.4 Computational Cost and What Was Measured

For $n = |U_q|$, embedding dimension d , maximum depth H , and B leaf balls, dense scoring and sorting cost $O(nd + n \log n)$. Recursive assignment costs $O(Hnd)$ because every level processes at most all candidates. Facet scoring and ordering contribute $O(n + Bd + B \log B)$ apart from tokenization; bounded insertion is linear in the traversed candidate list. Working storage is $O(nd + n + B)$. These bounds exclude encoder and generator inference, which have different computational profiles.

The native confirmation efficiency record reports 8.544 seconds for retrieval reconstruction and a 670,420,228-byte embedding cache. Generation totals are 887.650 seconds for BGE, 884.198 for Protected, 886.058 for NoFacet, and 888.282 for Unprotected. Peak GPU memory is 3.592 GiB. These measurements describe a cached local transaction; they are not full index-construction costs, uncached retrieval latency, or evidence of a stable speed advantage from a few seconds of timing difference.

Cache identity and deterministic subsets help interpret execution records, but cache hits are not independent generated-answer reproductions. The historical full or pre-selected subset rerun checks remain documented in their original records. The present writing task does not rerun those checks, infer new failure rates, or benchmark a repaired implementation.

8 Discussion

8.1 What the Compact Gain Establishes

The repeated compact differences demonstrate that an inference-time completion layer can improve the tested MiniLM system without replacing its encoder. The facet contrast further favors query-conditioned selection over the specific centroid-only alternative. This is a meaningful result about a constrained evidence-selection problem: a flat relevance list can leave recoverable information outside the final context.

The comparison does not determine why granular-ball organization is the best way to recover that information. Ordinary grouping, relevance–diversity selection, and direct unit-level coverage could account for some or all of the increment. The scalar compactness properties do not settle that question, because they hold for arbitrary partitions. The appropriate interpretation is therefore a useful complete pipeline plus limited component evidence, rather than proof that every structural component is irreducible.

8.2 Interpreting the Strong-Backbone Boundary

The BGE results change the practical claim. A user choosing between original compact Full and BGE should not infer superiority from the MiniLM-relative gain: the direct comparison favors BGE. The sidecar and native studies ask whether structure can then complement that stronger reference. Their intervals do

not establish the desired increment.

One explanation is that BGE already retrieves much of the available useful evidence, leaving fewer opportunities for completion. Other explanations are poor candidate scoring, overly restrictive gates, insufficiently informative lexical facets, or a mismatch between the selector and generator. The existing comparisons do not identify which explanation dominates. Because both algorithm definitions and query sets vary, a universal law that benefit decreases with retriever strength would exceed the evidence.

The negative original-versus-BGE result and unresolved extension results also have different meanings. The former interval is wholly negative; the latter permit small effects in either direction. Neither should be hidden behind a single qualitative success/failure label.

8.3 Selection and Placement Are Distinct Decisions

Protected placement is preferable to front insertion in the cross-space study, yet Protected does not improve on BGE. Thus, placement can make a candidate intervention less damaging without validating the candidate set. The native protection contrast remains unresolved, showing that the placement finding is not automatically portable between representations.

A pure position experiment would retain the same final evidence texts and verify their identity after token truncation, then change only their order. A pure evidence-set experiment would fix the placement rule while changing the selected units. The historical insertion-set comparison does not automatically satisfy the first condition. Maintaining these distinctions prevents a policy effect from being described as an isolated attention or position mechanism.

8.4 Implications for Future Matched Comparisons

The smallest informative extension would compare completion rules on a fresh compact boundary while matching candidate access, final K , protected prefix, changed positions, genuinely new units, and token accounting. Ordinary clusters should match the actual group count of each candidate configuration; a random grouping can additionally match group-size distribution. Pure component ablations should zero the designated coefficient while leaving the others unchanged. Independently tuned pipelines answer a separate question and should be reported separately.

Such a study would need a prospective development/confirmation split, a minimum meaningful effect, a finite tuning budget, and stopping rules. Formal uncertainty should follow a justified sampling model and null hypothesis. These are requirements for a future experiment, not additional results of this article. Corpus-scale retrieval and external structured methods would additionally require aligned corpora, readers, and budgets.

9 Limitations

The evidence is restricted to closed candidates in two English multi-hop QA benchmarks. We do not evaluate corpus reachability, ANN recall, index storage, full-wiki answer quality, multilingual transfer, or unanswerable questions. MuSiQue’s paragraph-level support annotations cannot be reinterpreted as sentence-level reasoning-chain labels. Query terms also need not contain the bridge entity required for a multi-hop answer.

Only one compact and one strong backbone are evaluated, with one principal generator and one additional deployment configuration. The results cannot establish broad generator robustness or a continuous relation between retriever capacity and completion utility. Training-release questions may have appeared in generator pretraining; project-held-out selection does not eliminate that possibility.

Several mechanism claims remain unidentified. The compact radius split trigger is redundant, the selector is static, group facets may exceed selected-unit facets, and the native NoFacet comparator changes

multiple weights. There is no completed matched flat-unit, ordinary-clustering, MMR, or sentence-coverage experiment. The mathematical coverage bound belongs to a distinct surrogate and supplies no guarantee for historical answer F1.

The legacy scoring discrepancy has not been quantified on the actual predictions. Historical tail-probability calibration and source-document dependence limit inferential claims. We do not silently replace the historical statistics with a new test. The confidence intervals are not equivalence bounds or uncertainty over model-development decisions. Mechanism audits and qualitative examples are post-decision descriptions, not new causal tests.

10 Conclusion

HyperGranular-RAG frames structured evidence completion as a bounded intervention over a dense retrieval ranking. The historical compact implementation repeatedly improves MiniLM answer F1 in the evaluated closed-candidate settings, and facet-conditioned selection exceeds its specified compact centroid-only control. The original system nevertheless falls below BGE, and the tested cross-space and native extensions do not establish an increment over BGE. Protected placement reduces harm relative to front insertion in one cross-space setting without establishing the usefulness of the intervention against the untouched baseline.

The analysis explains why these outcomes are compatible. Group compactness, lexical coverage, candidate addition, prompt-visible evidence, and answer quality are different quantities. A method can improve one without improving the next. Precise algorithm versions and matched comparisons are therefore essential to attributing a gain. The existing evidence supports completion for the tested compact system and motivates a narrower, testable question about which candidate interventions can complement strong retrieval.

Ethics, Availability, and Research Responsibility

The study uses existing benchmark releases and documented model artifacts; no new participant study is introduced. Positive, negative, and unresolved results are retained. Third-party data and weights remain governed by their original terms. The research repository contains implementations, frozen derived records, and source data for the reported figures. An anonymous artifact distribution and final reuse-license statement remain matters for the human authors before submission.

AI tools assisted software work, analysis preparation, and manuscript writing. They are not authors and do not bear responsibility for the work. Human authors must confirm authorship, affiliations, funding, conflicts, and the final venue-specific disclosure. No unconfirmed declaration is inferred here.

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A Historical Evidence Mapping

The paper’s scientific comparisons correspond to the following internal records: initial compact HotpotQA and MuSiQue are Stage4E and Stage4F; generator transfer is Stage4G; joint components are Stage4H; the cross-space sidecar is Stage4I; and native confirmation is Stage5A. These identifiers are provenance labels, not the scientific organization of the argument. The shared manuscript ledger lists the exact aggregate paths, effect values, and historical-method interpretation.

The baseline repository commit for the preceding audit was:

8a8e7cef8551c14bcea99e6a41a2b5a83f65456f

The old PDF and source are retained separately from this manuscript version. Some aggregate files retain a pending-verification status because verification was written to a separate final artifact; the aggregation file is not rewritten to erase that transaction history.

Later Stage6 development artifacts, its partially prepared official pipeline, and the repaired v2 synthetic prototypes are not included in the empirical tables. Their mathematical counterexamples help delimit possible claims, but their synthetic behavior is not evidence about held-out answer quality. No restricted reservation or Stage3B data is used in this manuscript revision.

B Terminology and Reporting Units

Term	Meaning used in this article
Evidence completion	Bounded change to an existing evidence list; no promise of complete reasoning support.
Granular ball	Local embedding group with a computed center and radius; historical compact split eligibility is not independently radius-triggered.
Facet	Fixed-tokenizer query term occurring in a group’s title/sentence union, not a verified semantic subquestion.
Facet hyperedge	Seed-to-expansion group relation based on seed-relative facet novelty and frozen gates.
Protected prefix	Initial dense entries preserved in order by the placement rule.
Placed candidate	Candidate accepted into the insertion segment; may be a promotion from dense ranks 11–20 in the compact method.
Genuinely new unit	Unit in the completed list but absent from the baseline Top-20.
Evidence coverage	Coverage of annotated supporting identities at the dataset’s sentence or paragraph granularity.
Answer utility	Observed change in answer EM/F1, not a score used by the confirmation retriever.
Unresolved effect	Recorded interval crosses zero; not evidence of equivalence.
Legacy scoring	Frozen alias-max token-overlap evaluator used by component, sidecar, and native studies; lacks the HotpotQA special-answer F1 exception.